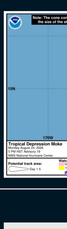
Stormdom

Global Storm Tracker

Tropical Storm LALA

Typhoon SAUDEL

Tropical Depression MOKE



Tropical Storm NARRA

Tropical Storm ISELLE

Graphics produced by Climate Prediction Center

See my new article: *How I Met Hurricane Camille*

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from the condensation of water vapor that has evaporated from tropical ocean surfaces.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (33-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes An Atlantic, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a Hurricane, in the Northwest Pacific (west of the International Dateline) a Typhoon, in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a Severe Tropical Cyclone, in the North Indian Ocean a Severe Cyclonic Storm, and in the Southwest Indian Ocean a Severe Tropical Storm.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down. Considerable flooding.

Category 3 – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend far inland.

Category 5 – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

SAFFIR-SIMPSON SCALE

Category	Knots	MPH	km/h	Damage
1	64-82	74-95	119-153	Minimal
2	83-95	96-110	154-177	Moderate
3	96-112	111-129	178-208	Extensive
4	113-136	130-156	209-251	Extreme
Super Typhoon	130+	150+	241+	Catastrophic
5	137+	157+	252+	Catastrophic

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by the width of the water depth increases offshore. In deep-water areas, such as the open ocean, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left front in the Southern Hemisphere) of the storm as it landfalls. The greatest surge is where winds are from the edge of the eyewall. Contrary to the popular myth, the storm surge is entirely wind-driven water—it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge is much worse than an 18-foot surge at low tide.

Tropical Cyclone Formation

By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

In the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west northwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a "convergence", where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a sophisticated wind as the axis of the tropical wave approaches. Gusts up to 23 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be some to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.

When the wave passes over warmer waters (SST's), convection and resulting rainfall are enhanced. This greater rainfall is concentrated with cumulonimbus clouds and thunderstorms. At the same time pressures fall to 1010mb, it is likely that the northeast wind has closed off to a southwest wind on the backside of the wave. The forward motion of the wave has also increased as much as 5 mph over the ocean waters. The wave is a broad low-level center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for further development. There are two conditions that must be present for the tropical depression to continue its upward development. The first condition is the ocean waters must be at least 79° Fahrenheit (26° Celsius) and low vertical shear. A tropical cyclone derives its power from the warm waters below. In addition, a strong anticyclone directly above the low-level inflow is essential. The edge of the hurricane's upper-level outflow, and may have to contend with both subsidence and shear. The effect of shear on a cyclone can range from a failure to develop further to catastrophic collapse of the tropical cyclone's support structure.

Tropical cyclones have the low-level circulation and the upper-level circulation (outflow), which a formation was discussed above. There is also a mid-level circulation. The mid-level circulation is similar in structure to the low-level circulation, and critical to the survival of a tropical cyclone that is passing over land. The mid-level circulation can be severely disrupted, or even dissipated, by interaction with land, especially mountainous terrain. The mid-level circulation remains robust if the cyclone moves over flat terrain, or over warm ocean waters, providing other factors are favorable.

Assuming all the ingredients are in place—warm SSTs, upper-level high pressure, and falling surface pressures—the cyclone will develop and reach a point of rapid intensification. It is one of nature's perfect machines. As warm waters feed the convection swirling around the center, heavy rainfall lowers surface pressures, high pressure aloft evacuates the inflow, which intensifies the inflow of warm, moist air, which in turn increases the rainfall and brings down the surface pressure even more.

Eye formation begins when a tropical storm reaches approximately 65mph, provided conditions are favorable for strengthening to continue. The eyewall begins to make its appearance, usually on the southwest quadrant. Hemispheric variations aside, the eye is usually the last feature to form. The stronger the cyclone becomes, the smaller and more circular the eye becomes. Eventually, annular hurricanes develop, which have a large, circular eye.

The eyewall replacement cycle is a great deal of durability provided that the upper level support remains and the southerly (Northern Hemisphere) vertical inflow persists. Here are some examples. In 1998, Hurricane Mitch developed into a supermassive Category 5 hurricane over the Caribbean Sea, weakening began when the center moved to a position directly north of Honduras, a country on the southern inflow, even though the storm was still over warm water. Mitch maintained its mid-level circulation as it drifted southward and inland across Central America because he was able to advect moisture from the East Pacific. Reintensification to tropical storm strength was almost immediate after reaching the Bay of Campeche. In 1988 Gilbert hit the Yucatan near Cancun as a Category 5 hurricane. The Yucatan peninsula, though flat, extended far enough south through the critical southerly inflow zone that Gilbert never recovered, even after moving over the Bay of Campeche. Hurricane Ivan stalled of the southern Florida coast long enough that the core of the hurricane collapsed. Tropical Storm Elena and beyond, where the proximity of the eye to land is not necessarily a weakening factor, although the extensive shelf water can sometimes be a moderating factor, if a slow-moving storm generates enough upwelling. A recent southern Hemisphere example of this is Severe Tropical Cyclone Monica, which developed into a "super cyclone" just north of Australia. The northerly (equatorward) inflow over the Arafura Sea fueled this storm.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, and finally Landfall.

Upwelling. When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the wave action caused by the strong winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface over an area 200 to 300 miles across. The result is weakening. It is possible for a hurricane to stall in one area long enough that it deteriorates. In 2004, Hurricane Frances stalled of the southeast Florida coast long enough that the core of the hurricane collapsed. Frances eventually made landfall as a Category 2 hurricane, instead of the earlier expected Category 3.

Entraining dry air. Sometimes, usually the post-season, when cyclones approach continental land masses, they may entrain dry air as part of their interaction with frontal troughs that carry cool, dry air behind them. The dry air kills off the convection that drive the hurricane's engine. If the dry air entrains deeply enough, the result can be dissipation.

Moving over cool water. Similar to upwelling, when a tropical cyclone moves over cool water (below 77° Fahrenheit/25° Celsius), it begin to weaken. Eventually this causes dissipation, particularly in the East Pacific. In the Atlantic, if the storm is caught in the mid-latitude westerlies and begins to recurve to the northeast, it generally becomes an extratropical storm by the time it has reached about 45° N. This is the so-called "graveyard" of Atlantic hurricanes. The storm becomes extratropical when it has transitioned to a cold-core, baroclinically driven system, and eventually dissipates. Or it may be absorbed by a large cyclone. There have been times when a hurricane passes north of the Azores and hits the British Isles as a Force 8 or stronger gale, having maintained a recognizable inner core.

Exposure to upper-level westerlies. Frequently, the ridge of high pressure that keeps the hurricane heading toward the west erodes. In the process, the hurricane begins moving upper-level westerlies diving down and impinging on the northern edge of the hurricane's upper-level support structure. Poleward outflow is restricted. A hurricane can still maintain itself when outflow is restricted in one quadrant. If outflow becomes restricted, troughing can disrupt and destroy the inner core.

After landfall. Tropical cyclones cannot survive over land, once their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can project life-threatening hurricane force winds over two hundred miles inland. As the storm progresses inland, a circular region of unusually warm—sometimes warmest in fact. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969 the strongest hurricane ever to make landfall on the continental United States, with a pressure of 190 mph and gusts well over 200 mph, which roared up the Mississippi Valley and eventually stalled over Virginia, where it produced catastrophic tropical storm strength rainfalls. Perhaps, however, the best example of this is Tropical Storm Allison, which, reaching landfall as a tropical storm with winds of only 72 knots, moved slowly from Texas to Louisiana to Texas and back to Louisiana, where it finally dissipated. Allison produced more than 30 inches of rain in Houston from the remnants of this storm.

Finally after landfall, the cyclone's circulation becomes more asymmetrical. The strongest winds will be found in the right front quadrant (left front in the Southern Hemisphere) of the storm as it landfalls. This is because the forward motion is added to the counter-clockwise punch of the storm. Thus, the strongest winds are found in the strong onshore flow. This is where the maximum storm surge flooding and the greatest potential loss of life, will be experienced. After landfall, the circulation's motion over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm's path.

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